# Making only as much as we need

# PERCENTAGE YIELD

# and ATOM ECONOM

Most of the substances we use every day are made from RAW MATERIALS, often through complex chemical reactions.

**Reactants (raw materials)**

**Chemical reactions**

**Products**

ICI

actavis
creating value in pharmaceuticals
Pottington

gsk GlaxoSmithKline

Hoechst

PERRIGO
Braunton

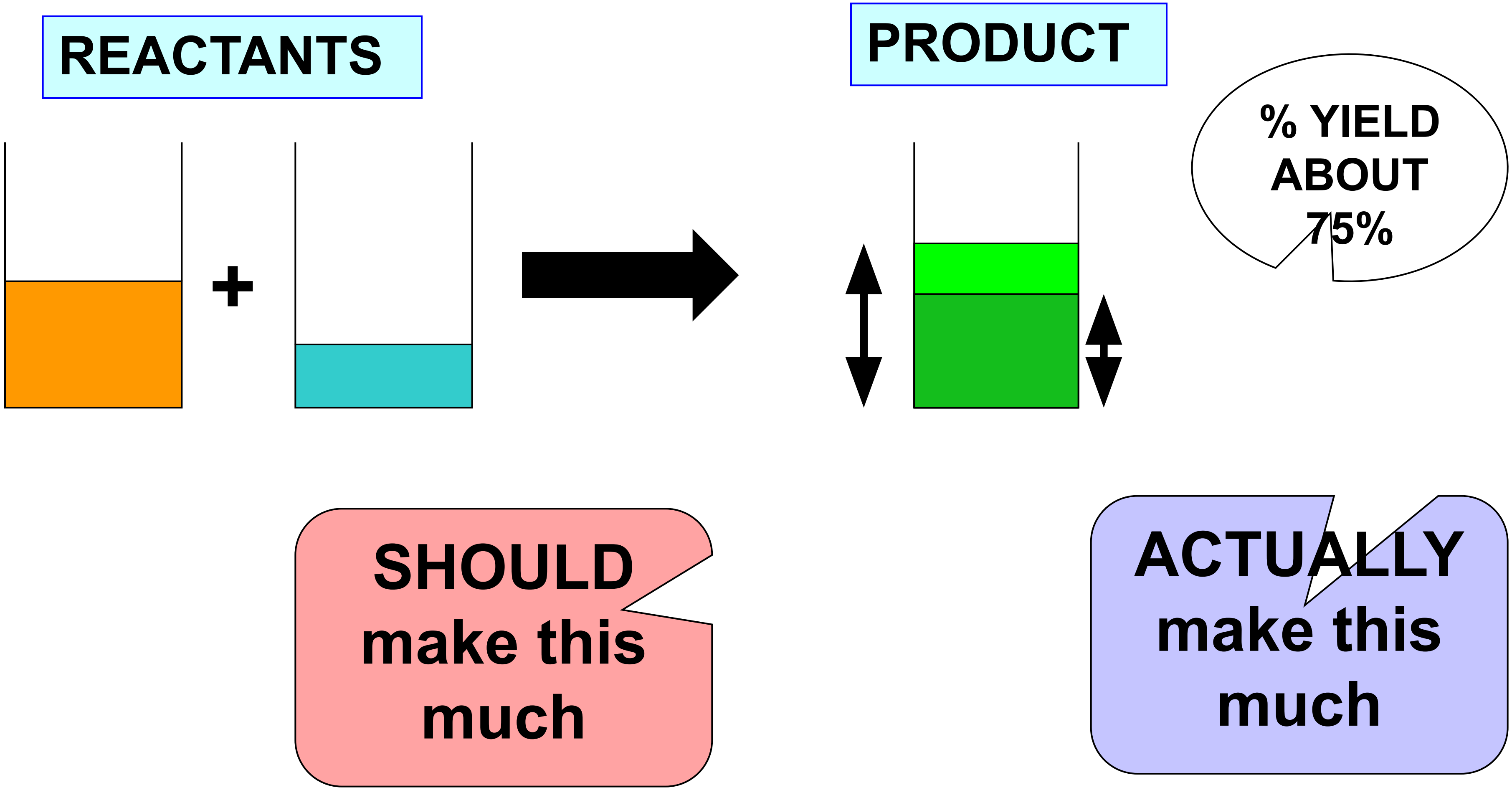
**The chemical industry is a multi billion pound international industry producing millions of products vital to our civilisation and well being.**

**Chemical Engineers play a crucial role and are much in demand – there are many opportunities and high levels of pay!**

**Chemical Engineers are much concerned with:**

**% YIELD and ATOM ECONOMY..**

**% YIELD** is the amount of product you **actually** make as a % of the amount you **should** theoretically make



# Old fashioned example: Cement from limestone

# PERCENTAGE YIELD 1

LIMESTONE (calcium carbonate) is used to make QUICKLIME (calcium oxide) for cement making

**RAM**
**Ca 40**
**O 16**
**C 12**

 → 

$$CaCO_3 = CaO + CO_2$$

**RFM:** **100** → **56** **44**

So, THEORETICALLY, **100 tonnes of limestone should produce 56 tonnes of quicklime.**

Why? – next slide

BUT the **ACTUAL** YIELD is only **48 tonnes**

**So..the PERCENTAGE YIELD is only** $\frac{48}{56} \times 100 =$ **87.5%**

# PERCENTAGE YIELD 2

**Very few chemical reactions have a yield of 100% because:**

- **The raw materials (eg limestone) may not be pure**
- **Some of the products may be left behind in the apparatus**
- **The reaction may not have completely finished**
- **Some reactants may give some unexpected products**

Careful planning and design of the equipment and reaction conditions can help keep % yield high

# Atom Economy

## Compare these two industrial reactions

$$2Mg + O_2 \rightarrow 2MgO$$

What do you notice about each one?

Think raw materials, useful products, waste products

USEFUL PRODUCT (antacids, fire resistant coatings, electrical insulators)

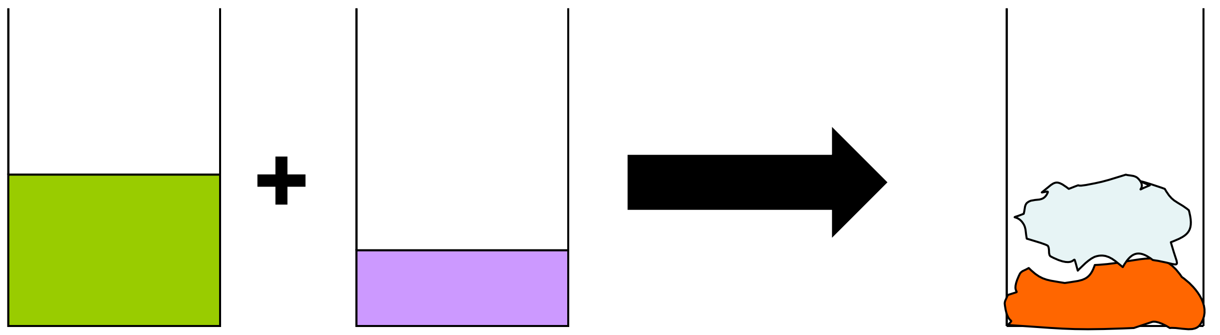
$$CaCO_3 \rightarrow CaO + CO_2$$

USEFUL PRODUCT (cement, glass, agriculture etc)

**ATOM ECONOMY** is the mass of the product **you want** as a % of the mass of **all** the products you make

**REACTANTS**

**PRODUCTS**



**Stuff you also get but don’t want**

**Stuff you want**

**ATOM ECONOMY about 50%**

# CALCULATING ATOM ECONOMY

Often, chemical reactions produce unwanted products along with the product you want.

**ATOM ECONOMY** is the mass of product you want as a % of the mass of all the products you make

Useful product

Waste product

$$CaCO_3 = CaO + CO_2$$

**RFM: 100 56 44**

$$\text{ATOM ECONOMY} = \frac{\text{mass useful product}}{\text{mass of all products}} \times 100\%$$

**Atom Economy** = 56 / (56 + 44) = 56 / 100 = **56 %**

**ATOM ECONOMY** is the mass of product you want as a % of the mass of all the products you make

**RAM**
**Mg 24**
**Fe 56**
**C 12**
**O 16**

$$2Mg + O_2 \rightarrow 2MgO$$

**RFM: 48 32 80**

**Atom Economy** = 80 / 80 x 100% = **<u>100 %</u>** **(obviously)**

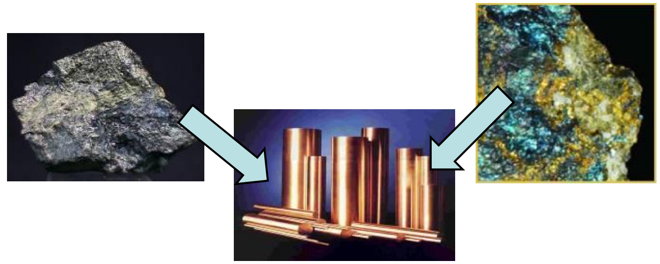
$$Fe_2O_3 + 3CO \rightarrow 2Fe + 3CO_2$$

$$160 \quad 84 \quad \underbrace{112 \quad 132}$$

**Atom Economy** = 112 / 244 x 100% = **<u>45.9 %</u>**

Find the atom economy for these 2 methods of extracting copper:

1. Heat copper oxide with carbon

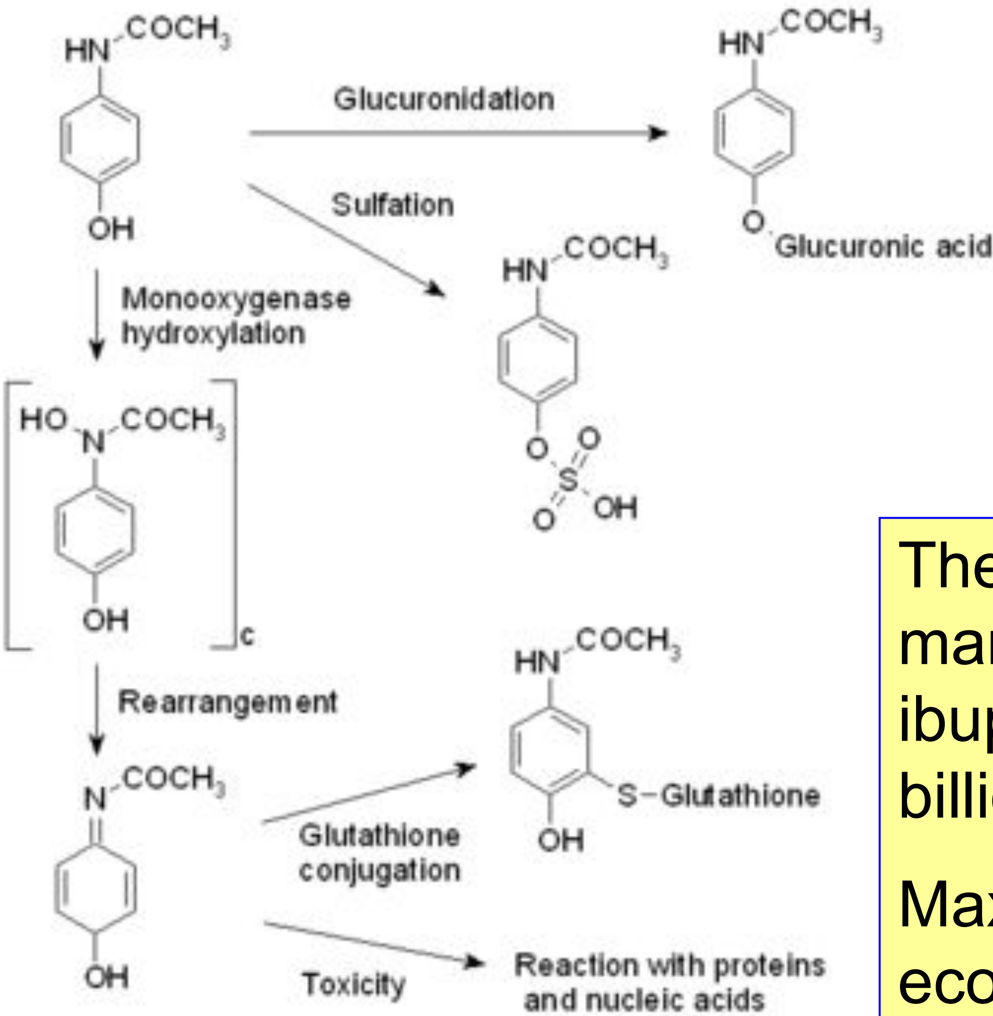


2. Heat copper sulphide with oxygen

**RAM Cu 64, O 16, C 12, S 32**

| Method 1 | Method 2 |
|---|---|
| $2CuO + C = \mathbf{2Cu} + CO_2$ | $CuS + O_2 = \mathbf{Cu} + SO_2$ |
| RFM **128** 44 | RFM **64** 64 |
| Mass of copper = 128 | Mass of copper = 64 |
| Mass of all products 128 + 44 = 172 | Mass of all products = 64 + 64 = 128 |
| **ATOM ECONOMY** $= \frac{128}{172} \times 100$ | **ATOM ECONOMY** $= \frac{64}{128} \times 100$ |
| **= 74.4 %** | **= 50 %** |

# Real example: Paracetamol



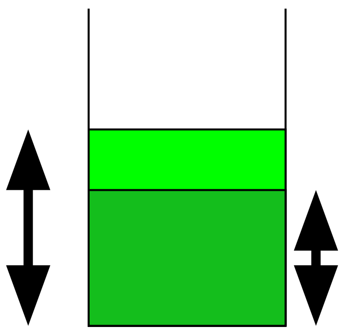

The non-prescription **analgesic** market (paracetamol, aspirin, ibuprofen) is worth about £21 billion annually.

Maximising % yield and atom economy in the reactions at left is vital to save money and conserv energy and resources

# SUMMARY

**% YIELD** is the amount of product you **actually** make as a % of the amount you **should** theoretically make

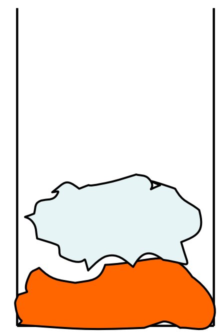
**ATOM ECONOMY** is the mass of the product **you want** as a % of the mass of **all** the products you make



**% YIELD ABOUT 75%**

**SHOULD make this much**

**ACTUALLY make this much**



**Stuff you also get but don't want**

**Stuff you want**

**ATOM ECONOMY about 50%**

Raw materials are scarce and expensive and so must be carefully conserved. Also, chemical processes need to produce as little waste as possible, minimise costs, energy use and pollution.

CHEMICAL ENGINEERS must plan to maximise:

PERCENTAGE YIELD by…

Using the most EFFICIENT REACTION CONDITIONS & APPARATUS to…

Reduce energy use, costs and conserve raw materials

ATOM ECONOMY by…

Choosing the most EFFICIENT REACTION to make the product to…

Reduce waste and pollution